Ex. No. 10	System Insight: Mastering Memory and Disk Resource Allocation
Date: 24.03.2025	

Objective:

To understand and analyse the concept of Mastering Memory and Disk Resource Allocation.

Theme: Online Progress Tracker

Description:

Efficient memory and disk management is vital for maintaining system performance. This theme explores how customer feedback data impacts memory, swap, and storage. Using monitoring scripts and allocation simulations, we analyze usage and optimize space. The goal is to ensure smooth processing of feedback through smart resource handling.

Exercise:

Question 1: Display Memory and Disk Usage Statistics.

```
 GNU nano 6.0                                  e101.c *
#include <stdio.h>
#include <sys/statvfs.h>

int main() {
    struct statvfs stat;
    if (statvfs("./theme_files", &stat) != 0) {
        perror("statvfs failed");
        return 1;
    }

    unsigned long total = stat.f_blocks * stat.f_frsize;
    unsigned long free = stat.f_bfree * stat.f_frsize;

    printf("Total space: %lu bytes\n", total);
    printf("Free space: %lu bytes\n", free);
    return 0;
}
```

```
kali% gcc ex101.c -o ex101 -lrt
kali% ./ex101
Displaying disk usage for tracker_data/progress.txt:
0       tracker_data/progress.txt
kali%
```

Question 2: Automate Cleanup of Unnecessary Files.

```
 GNU nano 6.0                                    ex102.c *
#include <stdio.h>
#include <stdlib.h>
#include <sys/stat.h>

int main() {
    mkdir("tracker_data", 0777); // Ensure dir exists

    printf("Cleaning up .tmp files in tracker_data\n");
    system("find tracker_data -name \"*.tmp\" -type f -delete");

    return 0;
}
```

```
kali% gcc ex102.c -o ex102 -lrt
kali% ./ex102
Cleaning up .tmp files in tracker_data
kali%
```

Question 3: Simulate File Fragmentation and Defragmentation.

```
File  Actions  Edit  View  Help
 GNU nano 6.0                                    ex103.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main() {
    printf("Monitoring memory usage (press Ctrl+C to stop)...\n");
    while (1) {
        system("free -h");
        sleep(2);
    }
    return 0;
}
```

```
Monitoring memory usage (press Ctrl+C to stop)...
               total        used        free      shared  buff/cache   available
Mem:           3.8Gi       783Mi       2.9Gi       9.3Mi       449Mi       3.1Gi
Swap:          974Mi          0B       974Mi
               total        used        free      shared  buff/cache   available
Mem:           3.8Gi       783Mi       2.9Gi       9.3Mi       449Mi       3.1Gi
Swap:          974Mi          0B       974Mi
               total        used        free      shared  buff/cache   available
Mem:           3.8Gi       783Mi       2.9Gi       9.3Mi       449Mi       3.1Gi
Swap:          974Mi          0B       974Mi
               total        used        free      shared  buff/cache   available
Mem:           3.8Gi       783Mi       2.9Gi       9.3Mi       449Mi       3.1Gi
```

Question 4: Monitor Memory Consumption.

```
 GNU nano 6.0                                        ex104.c *
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
#include <sys/stat.h>

int main() {
    mkdir("tracker_logs", 0777);

    FILE *fp = fopen("tracker_logs/disk_usage.log", "a+");
    if (fp == NULL) {
        perror("Could not open log file");
        return 1;
    }

    time_t now = time(NULL);
    fprintf(fp, "\n[Log Time: %s]\n", ctime(&now));
    system("df -h >> tracker_logs/disk_usage.log");

    fclose(fp);
    printf("Disk usage logged!\n");
    return 0;
}
```

```
kali% nano ex104.c
kali% gcc ex104.c -o ex104 -lrt
kali% ./ex104
Disk usage logged!
kali%
```

Question 5: Simulate Memory Paging

```
 GNU nano 6.0                                        ex105.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/stat.h>

int main() {
    mkdir("tracker_data", 0777);
    FILE *fp = fopen("tracker_data/download.txt", "w");
    if (fp == NULL) {
        perror("Failed to create file");
        return 1;
    }

    char data[1024];
    memset(data, 'A', sizeof(data));
    fwrite(data, sizeof(data), 1, fp);
    fclose(fp);

    printf("Simulated a 1KB file download to tracker_data/download.txt\n");
    return 0;
}
```

```
kali% nano ex105.c
kali% gcc ex105.c -o ex105 -lrt
kali% ./ex105
Simulated a 1KB file download to tracker_data/download.txt
kali%
```

Question 6: Calculate Space Occupied.

```
 GNU nano 6.0                                    ex106.c *
#include <stdio.h>
#include <stdlib.h>
#include <sys/stat.h>

int main() {
    mkdir("tracker_logs", 0777);

    FILE *fp = fopen("tracker_logs/mem_log.txt", "a+");
    if (fp == NULL) {
        perror("Could not open log file");
        return 1;
    }

    fprintf(fp, "\nMemory Usage Snapshot:\n");
    system("free -h >> tracker_logs/mem_log.txt");

    fclose(fp);
    printf("Memory usage snapshot saved.\n");
    return 0;
}
```

```
kali% nano ex106.c
kali% gcc ex106.c -o ex106 -lrt
kali% ./ex106
Memory usage snapshot saved.
kali%
```

Question 7: Simulate Virtual Memory Allocation.

```
kali% nano ex107.c
kali% gcc ex107.c -o ex107 -lrt
kali% ./ex107
Dummy progress files created!
kali%
```

```
 GNU nano 6.0                                    ex107.c *
#include <stdio.h>
#include <stdlib.h>
#include <sys/stat.h>

int main() {
    mkdir("tracker_data", 0777);

    const char *courses[] = {"math", "science", "history", "cs"};
    for (int i = 0; i < 4; i++) {
        char filename[100];
        snprintf(filename, sizeof(filename), "tracker_data/%s_progress.txt", cours

        FILE *fp = fopen(filename, "w");
        if (fp == NULL) {
            perror("error creating progress file");
            continue;
        }

        fprintf(fp, "Progress: 0%%\n");
        fclose(fp);
    }

    printf("Dummy progress files created!\n");
    return 0;
}
```

Question 8: Generate Disk Usage Reports.

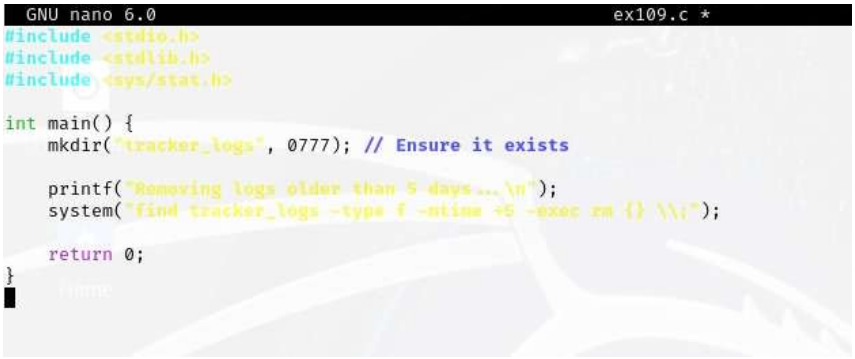
```
 GNU nano 6.0                                    ex108.c *
#include <stdio.h>
#include <stdlib.h>

int main() {
    printf("Checking available disk space:\n");
    system("df -h");

    return 0;
}
```

```
Checking available disk space:
Filesystem      Size  Used Avail Use% Mounted on
udev            1.9G     0  1.9G   0% /dev
tmpfs           393M  944K  393M   1% /run
/dev/sda1        78G   14G   60G  19% /
tmpfs           2.0G     0  2.0G   0% /dev/shm
tmpfs           5.0M     0  5.0M   0% /run/lock
tmpfs           2.0G  764K  2.0G   1% /tmp
tmpfs           393M  100K  393M   1% /run/user/1000
kali%
```

Question 9: Monitor Swap Usage.

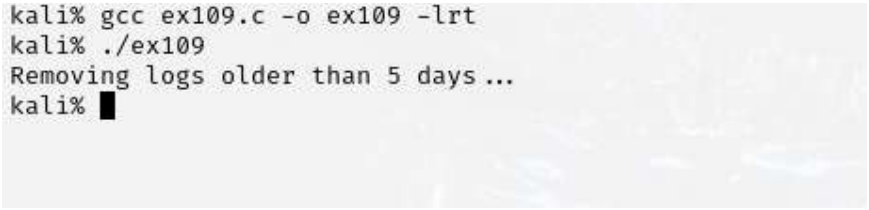


```
 GNU nano 6.0                                    ex109.c *
#include <stdio.h>
#include <stdlib.h>
#include <sys/stat.h>

int main() {
    mkdir("tracker_logs", 0777); // Ensure it exists

    printf("Removing logs older than 5 days...\n");
    system("find tracker_logs -type f -mtime +5 -exec rm {} \\;");

    return 0;
}
```



```
kali% gcc ex109.c -o ex109 -lrt
kali% ./ex109
Removing logs older than 5 days...
kali%
```

Question 10: Backup and Verify Data.



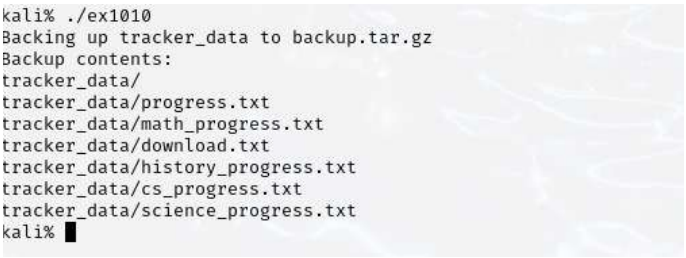
```
 GNU nano 6.0                                    ex1010.c *
#include <stdio.h>
#include <stdlib.h>
#include <sys/stat.h>

int main() {
    mkdir("tracker_data", 0777);
    const char *backup = "backup.tar.gz";

    printf("Backing up tracker_data to %s\n", backup);
    system("tar -czf backup.tar.gz tracker_data");

    printf("Backup contents:\n");
    system("tar -tf backup.tar.gz");

    return 0;
}
```



```
kali% ./ex1010
Backing up tracker_data to backup.tar.gz
Backup contents:
tracker_data/
tracker_data/progress.txt
tracker_data/math_progress.txt
tracker_data/download.txt
tracker_data/history_progress.txt
tracker_data/cs_progress.txt
tracker_data/science_progress.txt
kali%
```

Result:

The concept of Mastering Memory and Disk Resource Allocation in Unix and Linux has been done successfully.